Infinite NPA depth at minimal semidefinite complexity

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The quantum region of the CHSH value and the sum of two marginal averages requires infinite Navascués–Pironio–Acín depth, yet admits an exact finite semidefinite lift. We construct this lift by showing that every supporting bound is attained with equal measurement overlaps: one angle determines the entire boundary. Finite NPA certificates must remain valid for two independently varying angles, where the obstruction to finite depth survives. The same region has the smallest semidefinite extension degree possible for a nonpolyhedral set, separating infinite hierarchy depth from minimal semidefinite complexity.

Alice and Bob share a quantum state and each choose between two measurements with outcomes ± 1. Their observables are A_x, B_y, where $x, y \in \{0, 1\}$. Write $a_x = \langle A_x \rangle$, $b_y = \langle B_y \rangle$ and $E_{xy} = \langle A_x B_y \rangle$, and retain

$$M = a_0 + b_0, \qquad S = E_{00} + E_{01} + E_{10} - E_{11}. \quad (1)$$

S is the CHSH value [1]. Let $\mathcal{W}$ contain all pairs (M, S) attainable with quantum states and local measurements. The two marginal averages may differ; only their sum is retained. Shared randomness allows convex mixtures of strategies, so $\mathcal{W}$ is convex.

The preceding Letter [2] proved that every finite NPA level [3, 4] overestimates some supporting lines $S + \alpha M$ near $\alpha = 1$. Here we construct an exact finite lift of this same region: membership in $\mathcal{W}$ is equivalent to finitely many positive semidefinite (PSD) matrix constraints, with entries affine (constant plus linear) in M, S and auxiliary real variables. Write $\mathcal{W}_L$ for its level-L NPA projection. The semidefinite extension degree is the smallest possible largest matrix size in such a lift. We show

$$\mathcal{W} \subsetneq \mathcal{W}_L \quad \text{for every finite } L, \qquad \mathrm{sxdeg}(\mathcal{W}) = 2. \quad (2)$$

Our explicit lift uses matrices of sizes five and four. The optimal size two follows from a theorem on planar convex sets.

Equal overlaps determine every supporting line.— A supporting line bounds $hS + \alpha M$ for fixed real coefficients h, α. Define the Bell operator

$$C_{\mathrm{CHSH}} = A_0 B_0 + A_0 B_1 + A_1 B_0 - A_1 B_1,$$

where $A_x B_y$ means $A_x \otimes B_y$, and set $H = hC_{\mathrm{CHSH}} + \alpha(A_0 + B_0)$, with identities on the other party understood. The bound $hS + \alpha M \le \lambda$ holds in every state exactly when $\lambda I - H \succeq 0$; here $\succeq 0$ means PSD. Every binary measurement can be realized as a projective measurement on a larger space. Jordan's lemma then decomposes each local pair into blocks of dimension at most two. In each block choose axes so that

$$A_0 = B_0 = Z, \qquad A_1 = cZ + \sqrt{1 - c^2}X, \quad B_1 = dZ + \sqrt{1 - d^2}X. \quad (3)$$

Here X, Z are Pauli matrices and $c, d \in [-1, 1]$ are the cosines of the angles between each party's two measurement directions. These are the measurement overlaps. Deterministic strategies attain the local bound $\beta = 2|h| + 2|\alpha|$, so $\lambda \ge \beta$ is necessary.

For $\lambda \ge \beta$, a nonnegative determinant is enough to show that $\lambda I - H$ is PSD. Put $\Omega = 1 + c + d - cd \in [-2, 2]$ and expand $\det(rI + \lambda I - H) = r^4 + e_1 r^3 + e_2 r^2 + e_3 r + D$:

$$e_1 = 4\lambda, \qquad e_2 = 6\lambda^2 - 4\alpha^2 - 8h^2, \qquad e_3 = 4\lambda^3 - 8\lambda\alpha^2 - 16\lambda h^2 - 8h\alpha^2\Omega. \quad (4)$$

These three coefficients are nonnegative. For e_3, replacing its last term by $-16|h|\alpha^2$ gives an increasing cubic on $\lambda \ge \beta$, whose value at β is $16|\alpha|(|\alpha| + 2|h|)^2$. If $D \ge 0$ too, the polynomial is positive for every $r > 0$. A negative eigenvalue $-r$ of $\lambda I - H$ would make it zero, a contradiction.

It remains to compare the determinants for different measurements. Set $x = (1 - c)/2$, $y = (1 - d)/2$, and write $D(x, y) = \det(\lambda I - H)$. Direct expansion gives

$$D(x, y) = F(xy) + 64h^2(4h^2 - \alpha^2)xy(x + y), \quad (5)$$

where F is a polynomial in xy. For $|\alpha| \le |h|$, the coefficient of the sum is nonnegative. At fixed product, $x + y \ge 2\sqrt{xy}$, so

$$D(x, y) \ge D(\sqrt{xy}, \sqrt{xy}). \quad (6)$$

The equal-overlap point remains in $[0, 1]^2$. If the bound holds at every equal-overlap pair, this comparison proves it at every pair. Equal overlaps therefore attain the maximum.

For $|\alpha| \ge |h|$, the maximum is the local bound β. The endpoint tilted-CHSH inequality follows from the identity

$$4I - (C_{\mathrm{CHSH}} + 2A_0) = (I - C - A_1 D_B)^2 + \tfrac{1}{4}\sum_{y=0}^{1}[(I - A_0)(I + B_y)]^2, \quad (7)$$

where $C = (B_0 + B_1)/2$ and $D_B = (B_0 - B_1)/2$. Party exchange and output flips give either sign of CHSH and

either sign of the marginal. Average the Alice and Bob bounds, multiply by $|h|$, and use $\pm(A_0+B_0)\le 2I$ for the remaining weight $|\alpha|-|h|$. This proves $H\le\beta I$, attained by deterministic measurements with equal overlaps. The reduction thus holds for every real h,α.

One angle gives one finite lift.— At $c=d$, H preserves the singlet $|\psi_-\rangle=(|01\rangle-|10\rangle)/\sqrt{2}$, with eigenvalue $-2hc$. Use the ordered basis $(|\phi_+\rangle,|\phi_-\rangle,|\psi_+\rangle)$ for the triplet subspace, where $|\phi_\pm\rangle=(|00\rangle\pm|11\rangle)/\sqrt{2}$ and $|\psi_+\rangle=(|01\rangle+|10\rangle)/\sqrt{2}$. Set $c=(1-t^2)/(1+t^2)$, initially with $t\ge 0$. Multiplying this block of $\lambda I-H$ by $(1+t^2)^2$ removes the denominators and gives

$$K(t)=\begin{pmatrix}\lambda p_0-2hp_1 & -2\alpha p_0 & 0\\ -2\alpha p_0 & \lambda p_0-2hp_2 & -8ht^3\\ 0 & -8ht^3 & \lambda p_0+2hp_2\end{pmatrix}, \quad (8)$$

with $p_0=(1+t^2)^2$, $p_1=1-t^4$ and $p_2=1+4t^2-t^4$. At $t=0$ and in the leading coefficient, positivity forces $\lambda\ge 2h+2|\alpha|$ and $\lambda\ge -2h+2|\alpha|$. Together they imply $\lambda\ge\beta$, so the singlet bound holds too. Continuity includes $c=-1$, while $K(-t)=RK(t)R$, $R=\mathrm{diag}(1,1,-1)$. Consequently, for all real support coefficients,

$$hS+\alpha M\le\lambda \text{ on } \mathcal{W} \iff K(t)\succeq 0 \quad \forall t\in\mathbb{R}. \quad (9)$$

Positivity for all t has a finite description. A real matrix polynomial in one variable is PSD everywhere exactly when $K(t)=\sum_j Q_j(t)^TQ_j(t)$ for polynomial matrices Q_j, including singular cases [6]. Each diagonal entry of K is a sum of real squares of degree at most four, so every entry of Q_j has degree at most two: higher leading powers cannot cancel.

For a real vector $\xi=(\xi_1,\xi_2,\xi_3)^T$, the scalar $\xi^TK(t)\xi$ is therefore a sum of squares of linear combinations of the nine monomials $t^k\xi_i$, $0\le k\le 2$. Equivalently, it is a quadratic form in those monomials with a PSD coefficient matrix, called a Gram matrix. Average this representation under $(t,\xi)\mapsto(-t,R\xi)$. Terms that change sign under this transformation cancel, leaving two groups of monomials:

$$Z_+=(\xi_1,\xi_2,t\xi_3,t^2\xi_1,t^2\xi_2)^T,$$
$$Z_-=(\xi_3,t\xi_1,t\xi_2,t^2\xi_3)^T.$$

Equation (9) is therefore equivalent to the polynomial identity

$$\xi^TK(t)\xi=Z_+^TGZ_++Z_-^TJZ_-, \qquad G,J\succeq 0. \quad (10)$$

We now use the same monomials to test whether a pair (M,S) belongs to $\mathcal{W}$. Replace evaluation of polynomials by a linear functional. The entries of $Z_+Z_+^T$ and $Z_-Z_-^T$ span a sixteen-dimensional polynomial space. Let $\mathcal{L}_z$ assign real values $z\in\mathbb{R}^{16}$ to a basis of that space, extend it linearly, and apply it entrywise to set

$$U(z)=\mathcal{L}_z(Z_+Z_+^T), \qquad V(z)=\mathcal{L}_z(Z_-Z_-^T).$$

Define the three linear coordinate maps by

$$\mathcal{L}_z(\xi^TK(t)\xi)=\lambda\ell(z)-\alpha m(z)-hs(z).$$

If $U(z),V(z)\succeq 0$ and $\ell(z)=1$, applying $\mathcal{L}_z$ to Eq. (10) gives

$$\lambda-\alpha m(z)-hs(z)=\mathrm{tr}(UG)+\mathrm{tr}(VJ)\ge 0. \quad (11)$$

Thus $(m(z),s(z))$ satisfies every supporting inequality of $\mathcal{W}$.

Conversely, each real unit triplet state ξ at angle t supplies the functional $P\mapsto P(t,\xi)/(1+t^2)^2$. It gives

$$U=\frac{Z_+Z_+^T}{(1+t^2)^2}, \qquad V=\frac{Z_-Z_-^T}{(1+t^2)^2}, \quad (12)$$

with $\ell(z)=1$ and coordinates equal to the state's M,S. The scaled features have limits at $t=\infty$. The pairs (M,S) obtained this way form a compact set. Their convex hull has the same maxima of $hS+\alpha M$ as $\mathcal{W}$: equal overlaps suffice, and each singlet point $(0,-2c)$ is a mixture of the triplet endpoint points $(0,\pm 2)$. Their convex hull is therefore $\mathcal{W}$. Mixing the corresponding functionals gives

$$\mathcal{W}=\{(m(z),s(z)):\ell(z)=1,\ U(z),V(z)\succeq 0\}. \quad (13)$$

The complete proofs give every entry of U,V [5]. The set of triplet pairs is also connected. Every point in the convex hull of a compact connected planar set lies on a segment joining two points of the set [5]. Thus every (M,S) is a mixture of two real equal-overlap strategies. Each matrix is then a sum of two rank-one matrices, so U,V can simultaneously have rank at most two.

Infinite depth, minimal block order.— Equal overlaps attain the quantum maximum. Finite NPA certificates, however, must hold for independent measurement angles. Let $f(\alpha)=\max_{\mathcal{W}}(S+\alpha M)$ and let $f_L(\alpha)$ be its level-L NPA upper bound. The NPA error is f_L-f; the quantum violation is $f-(2+2|\alpha|)$. The preceding proof [2] shows that a bounded ratio of these quantities as $\alpha\uparrow 1$ would give a sum of polynomial squares for

$$\mathfrak{m}(X,Y)+c, \qquad \mathfrak{m}=1+X^4Y^2+X^2Y^4-3X^2Y^2, \quad c\ge 0.$$

Here X,Y are rescaled real angle variables. Such a sum of squares is impossible: constant values on both axes restrict each cubic square factor to $a+bXY+dX^2Y+eXY^2$, whose squared X^2Y^2 coefficient is nonnegative. Yet on equal angles,

$$\mathfrak{m}(X,X)=(X^2-1)^2(2X^2+1).$$

The two-angle polynomial is not a sum of squares, although its equal-angle restriction is. The complete rescaling and limit argument appears in the companion [5].

The quantum maximum satisfies [2, 7]

$$f(1-T)=4-2T+\tfrac{4}{3}T^3+O(T^4), \qquad T\downarrow 0.$$

For a polygon, this maximum would be affine in T sufficiently near zero. Thus $\mathcal{W}$ is not a polygon. PSD constraints are polynomial inequalities; their projections are semialgebraic, meaning finite unions of sets defined by polynomial equalities and inequalities. Scheiderer's theorem gives extension degree at most two for every closed convex semialgebraic set in the plane [8]. Scalar PSD constraints describe polyhedra, so size one cannot suffice. Hence $\mathrm{sxdeg}(\mathcal{W}) = 2$.

The construction also gives thirty-two exact projections. For each CHSH sign pattern $\chi_{ij} \in \{\pm 1\}$ with $\prod_{ij} \chi_{ij} = -1$, and each selected input pair x, y, relabelling inputs and outputs identifies

$$\left(a_x + \chi_{xy} b_y,\ \sum_{ij} \chi_{ij} E_{ij} \right)$$

with the same region $\mathcal{W}$. Adding these lifts to any NPA level retains every quantum correlation and gives the exact maximum of every linear combination of the two coordinates on each plane. This removes the NPA error on all thirty-two planes.

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[5] See the accompanying Supplemental Material for the complete finite-lift proof, explicit membership matrices, all signed asymmetric support branches, physical relabellings and the full earlier NPA proof.

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